

Appendix B Chapter Exams

Chapter 1 Exam

1. Which of the following is an example of documenting?
 - (A) The medication aide (MA) submits a written request to her supervisor to change from afternoon to morning shifts.
 - (B) The resident keeps a list of her medications at her bedside.
 - (C) The MA administers eyedrops to a resident and records the event on the resident's chart.
 - (D) The nurse provides the MA with a list of medications for a new resident.
2. MAs are responsible for
 - (A) Administering prescription and nonprescription medications to residents according to physicians' orders and under a nurse's supervision
 - (B) Administering only nonprescription medications to residents according to physicians' orders and under a nurse's supervision
 - (C) Determining which residents need assistance taking medications and helping those residents
 - (D) Deciding the amount of medication to administer to a resident based on observation
3. Another name for a *medication aide* is
 - (A) Pharmacology technician
 - (B) Medication technician
 - (C) Medication coordinator
 - (D) Pharmacist's assistant
4. Providing person-centered care includes
 - (A) Providing medication updates to a resident's family members
 - (B) Supervising other MAs
 - (C) Treating all residents exactly the same
 - (D) Treating all residents as individuals
5. For an MA, behaving professionally involves
 - (A) Arriving on time and leaving early
 - (B) Delegating tasks to other members of the care team
 - (C) Doing only assigned tasks that the MA is trained to do
 - (D) Choosing which facility policies to follow
6. What should an MA do if she is assigned to perform a task she has not been trained to do?
 - (A) The MA should refuse the task and explain why she must refuse it.
 - (B) The MA should delegate the task to someone who knows how to perform it.
 - (C) The MA should take initiative and perform the task.
 - (D) The MA should ignore the assignment.
7. In which of the following cases might the MA be held liable for harming a resident?
 - (A) The resident experiences pain during the night after the MA would not provide a nonprescription painkiller requested by the resident. The medication was not ordered by a licensed provider.
 - (B) A resident's condition worsens after he refuses his evening medication. The MA had documented the refusal and reported it to the supervisor.
 - (C) The MA administers an established medication to the resident based on a medication order and the resident has an allergic reaction.
 - (D) The MA crushes a medication that is labeled *DO NOT CRUSH* and sprinkles it on applesauce to make it easier for the resident to swallow. The resident's heart rate and blood pressure drop to dangerously low levels.
8. When communicating with a resident who is difficult to understand the MA should
 - (A) Pretend she understands the resident to avoid conflict
 - (B) Take the time to listen to the resident, asking her to repeat herself if necessary
 - (C) Distract the resident with questions
 - (D) Complete the interaction as quickly as possible

9. How can an MA use effective communication skills to help promote resident safety?
- (A) The MA can listen to the resident's concerns and report her questions to the nurse.
 - (B) The MA can advise the physician about whether the resident's dosage is correct.
 - (C) The MA can pass along the resident's diagnoses to the resident's family members.
 - (D) The MA can inform the resident which side effects can be safely ignored.
10. Which of the following is true regarding the scope of practice of an MA?
- (A) Scope of practice in a facility is usually based on seniority.
 - (B) Scope of practice may not always be defined by law, but it is always defined within a facility.
 - (C) Scope of practice always includes diagnosing residents.
 - (D) Under scope-of-practice laws, nurses carry out instructions given by MAs.

Chapter 2 Exam

1. Which of the following is true regarding the difference between laws and ethics?
 - (A) Ethics are practiced by healthcare workers and laws are practiced by lawyers.
 - (B) Ethics guide a person to behave responsibly and laws enforce that behavior.
 - (C) *Laws* is simply another word for ethics.
 - (D) Ethics are harmful actions and laws are helpful actions.
2. If an MA realizes he has made a mistake, when is the best time to report it?
 - (A) Immediately
 - (B) At the MA's next performance evaluation
 - (C) Within 48 hours
 - (D) At the end of the MA's shift
3. The term *controlled substance* refers to
 - (A) Any medication administered by an MA
 - (B) A medication that is considered more likely to be abused
 - (C) A prescription medication
 - (D) Any medication not in pill form
4. Which of the following is the most important thing for an MA to know before performing an assigned task?
 - (A) How many residents will be taking nonprescription medications during her shift
 - (B) The exact number of residents to whom she will need to administer medications
 - (C) Whether the task assigned is within her scope of practice
 - (D) The approximate amount of time it will take to administer medications on a shift
5. An MA is never allowed to
 - (A) Split a pill
 - (B) Administer inhaled medications
 - (C) Receive medication orders by phone
 - (D) Refuse an assigned task
6. Which of the following is the most professional and respectful way for an MA to refuse a task they are not trained or allowed to perform?
 - (A) "You shouldn't be assigning this task to me."
 - (B) "This task is outside my scope of practice and I am not legally allowed to complete it."
 - (C) "Can't you assign this to someone else?"
 - (D) "Thank you for the assignment, but I will not be completing it."
7. The right to privacy and confidentiality
 - (A) Was taken away by HIPAA in 1996
 - (B) Is more about politeness than about legal rights
 - (C) Is about protecting long-term care facilities from lawsuits
 - (D) Ensures the resident's right to speak privately to anyone
8. In healthcare settings, *diversion* refers to
 - (A) Distracting a resident who is receiving an injected medication
 - (B) Theft of prescribed medications
 - (C) Dividing tasks among staff members
 - (D) A resident refusing a medication
9. What is an MA's role in preventing diversion?
 - (A) Refusing any tasks associated with counting or preparing medications
 - (B) Finding a secret hiding place for the key to medication storage
 - (C) Practicing careful and accurate documentation
 - (D) Delegating medication documentation to a nursing assistant
10. Which of the following is an example of negligence on the part of an MA?
 - (A) Not administering a medication when a resident refuses it
 - (B) Leaving a resident's medications on his bedside table because the resident is asleep
 - (C) Telling a resident the nurse has to approve a PRN medication before it can be administered
 - (D) Checking a resident's mouth to make sure a medication has been swallowed

Chapter 3 Exam

1. In pharmacology, the acronym *ADME* stands for
 - (A) Absorption, Development, Medication, Excretion
 - (B) Administration, Distribution, Medication, Excretion
 - (C) Absorption, Distribution, Metabolism, Excretion
 - (D) Administration, Distribution, Metabolism, Excrement
2. Which of the following statements is true of liquid medications?
 - (A) Only syrups are considered liquid medications.
 - (B) A tincture contains particles that have not completely dissolved.
 - (C) A spray is only used in the nose.
 - (D) They may be easier for residents to swallow than pills.
3. Which of the following are considered solid medications?
 - (A) Capsules, powders, and caplets
 - (B) Caplets, elixirs, and lozenges
 - (C) Liniments, lozenges, and solutions
 - (D) Tablets, suppositories, and suspensions
4. A controlled release medication cannot be _____.
 - (A) Swallowed
 - (B) Split or crushed
 - (C) A capsule
 - (D) Long-acting
5. Which of the following is true about medicated patches?
 - (A) They are administered under the tongue.
 - (B) They must be dissolved in alcohol.
 - (C) They prevent medication from entering the bloodstream.
 - (D) They are administered by sticking them to the skin.
6. Which of the following is true of generic medications?
 - (A) Generic medications are weaker than brand-name medications.
 - (B) Generic medications do not require a prescription and brand-name medications do.
 - (C) Generic medication names are always capitalized.
 - (D) Generic medication names are usually based on the medication's active ingredients.
7. The _____ route is the most common route of administration for medications.
 - (A) Injection
 - (B) Oral
 - (C) Optic
 - (D) Topical
8. The otic administration route is via the
 - (A) Ear
 - (B) Eye
 - (C) Nose
 - (D) Mouth
9. How might a resident's sex affect the way a medication works within that person's body?
 - (A) Men are likelier to have an adverse reaction to medication.
 - (B) Medication generally remains in a woman's body for a longer period of time.
 - (C) Medication generally remains in a man's body for a longer period of time.
 - (D) Sex has no effect on the way a medication works in a person's body.
10. The study of pharmacogenetics is the study of how a person's _____ affect(s) the way medications work in his body.
 - (A) Mobility
 - (B) BMI
 - (C) Liver
 - (D) Genes
11. MAs should be aware that certain _____ such as _____ may affect the body's response to some medications.
 - (A) Herbal supplements; grapefruit juice
 - (B) Vitamins; enzymes
 - (C) Foods; grapefruit
 - (D) Moods; boredom

12. MAs are permitted to administer medications via the following routes:
- (A) Vaginal, rectal, and intradermal
 - (B) Sublingual, transdermal, and ocular
 - (C) Rectal, topical, and intramuscular
 - (D) Oral, intravenous, and transdermal

Chapter 4 Exam

1. What is the difference between a prescription and a medication order?
 - (A) A medication order is for a generic medication and a prescription is for a brand name.
 - (B) A prescription is filled by a pharmacist and a medication order is for an OTC medication.
 - (C) A prescription medication is taken for a long period of time and a medication order is a one-time dose.
 - (D) Medication orders are filled in a healthcare setting while prescriptions are filled on an outpatient basis.
2. Which of the following providers is authorized to order medications?
 - (A) Registered Nurse (RN)
 - (B) Medication Aide (MA)
 - (C) Nurse Practitioner (NP)
 - (D) Licensed Practical/Vocational Nurse (LPN/LVN)
3. When an MA is given a *stat* medication order he should
 - (A) Administer the medication immediately
 - (B) Tell the practitioner that receiving medication orders is beyond his scope of practice
 - (C) Put the order in writing and give it to the nurse
 - (D) Call the resident's primary care doctor to verify the order
4. On a medication order the *strength* of a medication refers to
 - (A) The amount of time it requires to become effective
 - (B) The quantity of doses per order
 - (C) The amount of active ingredient in one unit
 - (D) The total amount of active ingredient per dose
5. Which of the following is true regarding the information contained in a MAR and an eMAR?
 - (A) A MAR is an internal document and an eMAR is a legal record.
 - (B) A MAR contains only information concerning current medications while an eMAR is a record of past medications.
 - (C) An eMAR is used in hospitals and a MAR is used in LTC facilities.
 - (D) An eMAR and a MAR contain the same information regarding a resident's medications.
6. When should the MA document the administration of a medication to a resident?
 - (A) Immediately after the medication is administered
 - (B) Thirty minutes after administration
 - (C) Twice per day, in the morning and evening
 - (D) At the beginning of the MA's shift
7. If a resident refuses a medication
 - (A) There is no need to document it
 - (B) The refusal must be documented
 - (C) The MA should insist that the resident take the medication
 - (D) The MA should call the resident's family to report the refusal
8. What is the most important reason medication documentation must be accurate?
 - (A) Because medications are expensive and the facility needs to know how much to order
 - (B) Because decisions about a resident's care are based on these records
 - (C) Because the MA's performance reviews and promotions are based on accurate documentation
 - (D) To report medication effectiveness back to medication manufacturers

9. Which of the following is an example of objective information?
 - (A) Ms. Fox was less cheerful than usual today.
 - (B) Ms. Fox did not sleep well last night and appeared tired all morning.
 - (C) Ms. Fox drank half a cup of tea (4 fluid ounces) and ate one bite of toast for breakfast.
 - (D) Ms. Fox has lost her appetite and appears depressed.
10. Which of the following events should be immediately reported to a supervisor?
 - (A) Swelling of the lips and tongue
 - (B) A blood pressure reading of 110/70
 - (C) A resident reports feeling cold
 - (D) A pulse rate of 82 beats per minute
11. Which of the following types of LTC facility document requires signatures or initials of more than one employee?
 - (A) A MAR
 - (B) A medication order for a controlled substance
 - (C) An eMAR with multiple medications listed
 - (D) A controlled substance log
12. If an MAR is missing any information the medication aide should
 - (A) Contact a supervisor prior to administering the medication
 - (B) Check the MAR of a different resident taking the same medication for the missing information
 - (C) Administer the medication as scheduled and flag it for the supervisor to fix
 - (D) Make the needed correction on the MAR
13. Why is it so important for the MA to ask a supervisor questions when a medication order is unclear or seems unsuited to a resident's needs?
 - (A) To help the MAs make better decisions about which medications to prescribe in the future
 - (B) To ensure the supervisor does not get into trouble with the resident's family members
 - (C) Because possible medication errors directly impact the resident's health and safety
 - (D) Because MAs have more time to research medications than nurses

Chapter 5 Exam

1. Why is it important for an MA to understand medication calculations and conversions even if she cannot perform them?
 - (A) Calculating and converting medications is not something an MA needs to understand.
 - (B) Understanding calculations and conversions may help the MA to be aware of potential errors.
 - (C) Understanding calculations and conversions allows the MA to quickly make substitutions and adjustments to residents' dosages.
 - (D) Understanding conversions and calculations helps an MA avoid wasting medications.
2. The most commonly used system of measurement in healthcare settings is
 - (A) The metric system
 - (B) The apothecaries' system
 - (C) The eMAR system
 - (D) The household system
3. Which of the following is equivalent to 0.5 g?
 - (A) 500 (gr)
 - (B) 5 mg
 - (C) 500 mg
 - (D) 50,000 mcg
4. In the metric system, liquids are measured in
 - (A) Micrograms
 - (B) Milliliters
 - (C) Milligrams
 - (D) Millimeters
5. In medicine, toxicity is
 - (A) A measurement of the active ingredient in a medication
 - (B) An overdose of a prescription medication
 - (C) A poisoning effect caused by a medication
 - (D) Caused when more than two medications are taken at one time
6. Household measurements are _____ than metric measurements.
 - (A) Less precise
 - (B) More precise
 - (C) The same as
 - (D) Smaller
7. There are _____ mL in a teaspoon
 - (A) 5
 - (B) 15
 - (C) 25
 - (D) 240
8. What does it mean when a medication has a narrow therapeutic index?
 - (A) The medication does not work on a majority of patients.
 - (B) The medication takes effect more quickly than most other medications.
 - (C) The medication does not stay in the body very long after it is administered.
 - (D) The maximum dosage in the dosage range is very close to the minimum therapeutic dose.
9. What is the most important reason not to use household spoons for liquid medication administration?
 - (A) There is wide variation in the size of household spoons, and this leads to inaccurate dosing.
 - (B) It is easy to spill medications being administered via a household spoon.
 - (C) The metal of household spoons can react with liquid medication ingredients.
 - (D) Residents with dementia may bite down on a spoon.
10. What is a clinical trial?
 - (A) The first prescription medication dose administered to a new resident
 - (B) The administration of medication in an inpatient setting
 - (C) The amount of time it takes a body to adjust to a new medication
 - (D) The monitored testing of a new medication to determine effectiveness

Chapter 6 Exam

1. The process by which a drug works by activating receptors is known as _____, and the process by which a drug works by deactivating receptors is known as _____.
 - (A) Antagonism; protagonists
 - (B) Agonism; antagonism
 - (C) Agism; agonism
 - (D) Absorption; antagonism
2. Which of the following statements about medication side effects is true?
 - (A) The number of side effects a medication can cause is always dependent on dosage.
 - (B) Side effects are completely unpredictable.
 - (C) Most side effects do not cause serious harm.
 - (D) Medications always cause the same side effects in all people.
3. An example of a systemic side effect is
 - (A) Injection site pain the day after a vaccination
 - (B) A dry mouth after using an inhaler
 - (C) Hypertension after long-term use of a corticosteroid
 - (D) Reduction of lung inflammation after using an inhaler
4. A side effect is a(n) _____ reaction to a medication, while an adverse drug reaction may be _____.
 - (A) Predictable; unpredictable
 - (B) Beneficial; unpredictable
 - (C) Allergic; expected
 - (D) Constant; localized
5. Which of the following is a category of adverse drug reaction (ADR)?
 - (A) Absorption reaction
 - (B) Paradoxical reaction
 - (C) Clinical reaction
 - (D) Insensitive reaction
6. Which of the following is a possible symptom of anaphylaxis?
 - (A) Itching that begins 1 day after an injection
 - (B) Trouble breathing that begins 5 minutes after medication ingestion
 - (C) Eye itching relieved by eye drops
 - (D) Chronic constipation
7. Drug tolerance occurs when
 - (A) A patient develops an addiction to the drug
 - (B) A drug has reached peak effectiveness
 - (C) Sudden hypersensitivity to a drug occurs
 - (D) The drug no longer produces the desired response
8. What is the MA's role related to a resident who may be developing drug tolerance?
 - (A) The MA should assess drug tolerance and begin to gradually reduce the resident's dosage.
 - (B) The MA should encourage the resident to occasionally refuse the medication to help prevent drug tolerance from occurring.
 - (C) The MA should report any observed changes in the resident's status.
 - (D) The MA should suggest substitution of a medication that treats the same condition.
9. Which of the following is true of drug-drug interactions?
 - (A) Some drug-drug interactions can produce a beneficial effect.
 - (B) All drug-drug interactions produce a beneficial effect.
 - (C) Drug-drug interactions only occur between prescription medications.
 - (D) Potentiation is a type of drug interaction that causes one drug to cancel out the beneficial effects of another drug.
10. Which of the following statements concerning herbal supplements is true?
 - (A) Residents are free to take any herbal supplements they wish, and they do not need to be documented.
 - (B) Herbal supplements do not interact with prescription or OTC medications.
 - (C) Herbal supplements slow absorption of most medications.
 - (D) Herbal supplements may interact with medications and a resident's doctor must know if they are taken.

Chapter 7 Exam

1. Which of the following is a “right” of medication administration?
 - (A) The right brand
 - (B) The right to confidentiality
 - (C) The right to safety
 - (D) The right time
2. In medication administration, the *right route* refers to
 - (A) The order in which different residents receive their medications
 - (B) The path the MA takes through the facility while distributing medications
 - (C) Getting a medication into or onto the resident’s body in the correct manner
 - (D) Receiving the medication order directly from the prescriber
3. When is the “right” time for the MA to administer medications?
 - (A) When it is convenient for the MA
 - (B) Within 30–60 minutes of the time indicated on the resident’s MAR
 - (C) Within 60 minutes of a meal
 - (D) Oral medications are administered in the morning and medications by other routes can be administered as needed throughout the day.
4. Which of the following would be considered a medication administration error?
 - (A) An MA allows the resident to take an oral medication with iced tea instead of water and does not document the beverage.
 - (B) The facility has replaced one generic medication with another generic of the same type from a different manufacturer.
 - (C) One resident receives an application of OTC muscle ointment intended for another resident with a similar last name.
 - (D) A resident receives his nighttime medications 15 minutes after the time specified on his MAR.
5. Medications should be administered along with food
 - (A) When indicated on the MAR
 - (B) At all times
 - (C) Only if the resident is hungry
 - (D) Only if the medication pass is occurring during a scheduled meal time
6. Prior to administering any medication, the MA must first establish the resident’s _____.
 - (A) Diagnosis
 - (B) Identity
 - (C) Meal schedule
 - (D) Religion
7. Which of the following is true about an MA’s role in holding medications?
 - (A) An MA should hold medication if the resident has recently vomited.
 - (B) An MA should hold a medication if the resident is napping.
 - (C) An MA should hold a medication to protect a resident’s privacy when she has a visitor.
 - (D) An MA should only hold a medication if it is indicated by the MAR.
8. Which of the following should an MA report to a nurse prior to administering a resident’s medication?
 - (A) Any change in the resident’s status or condition should be reported.
 - (B) The resident’s location in the facility should be reported.
 - (C) A blood pressure reading within the normal range for that resident should be reported.
 - (D) The time of the resident’s last meal should be reported.
9. What is a boxed warning?
 - (A) Instructions from the pharmacy that accompany a prescription medication
 - (B) A list of all known medication side effects
 - (C) A required warning related to serious or life-threatening medication side effects
 - (D) The controlled substance section of a resident’s MAR

10. Which of the following is true of an MA's duties as part of the care team?
- (A) The MA spends the most time with residents and is considered the "eyes and ears" of the care team.
 - (B) The MA cannot make decisions about a resident's medication but must report changes in the resident and any potential medication errors.
 - (C) The MA's role does not involve researching medications she administers.
 - (D) Because an MA reports directly to a supervising nurse, an MA is never responsible for medication administration errors.
11. An MA looking for reliable information about a drug he has never administered before should
- (A) Call a poison control helpline
 - (B) Look the drug up on Wikipedia
 - (C) Use the facility's approved drug reference
 - (D) Ask a more experienced MA
12. A contraindication is
- (A) A drug-drug interaction that may cause harm to a resident
 - (B) A reason a certain medication should not be taken by a particular resident or should be taken with caution
 - (C) A drug prescribed to a resident for a condition outside the usual use for that medication
 - (D) The total effect, positive or negative, produced by multiple medications within the body
13. Most medications should be stored in a _____ and kept away from _____.
- (A) Moist area; dry air
 - (B) Dry area; sunlight
 - (C) Cool area; freezers
 - (D) Dark area; other medications
14. When should a medication storage area be kept locked?
- (A) Any time it is not actively in use
 - (B) After all medications have been transferred to a medication cart
 - (C) When there is not a medication aide working in the facility
 - (D) At night
15. An MA can help promote safety in a facility by
- (A) Getting help with passing medications as often as possible
 - (B) Keeping keys to medication storage areas and carts secure
 - (C) Delegating a nursing assistant to help monitor unlocked medication carts
 - (D) Keeping all of a resident's medications in the resident's room
16. What should an MA know concerning medication disposal?
- (A) All unused medications should be flushed down the toilet at the end of an MA's shift.
 - (B) All disposal of medications must be documented regardless of reason.
 - (C) Only controlled substances require documentation for disposal.
 - (D) MAs always control all aspects of medication disposal for a facility.

Chapter 8 Exam

1. Unit dose packaging contains
 - (A) A single dose of a medication
 - (B) One day's worth of medication
 - (C) One week's worth of medication
 - (D) One month's worth of medication
2. Another common term for a *blister pack* is
 - (A) Stock medication pack
 - (B) Single resident pack
 - (C) Pharmacy pack
 - (D) Bubble pack
3. What is the most common form of prescription medication packaging in a long-term care facility?
 - (A) Unit dose packaging
 - (B) Medication cart packaging
 - (C) Blister packs
 - (D) Traditional packaging
4. A _____ is a mobile medication storage system from which an MA passes and documents medication administration.
 - (A) Drug station
 - (B) Medication cart
 - (C) Dispensary
 - (D) Charge station
5. An MA should perform hand hygiene _____ handling medications of any type and _____ residents.
 - (A) Before; between
 - (B) Before; after every two
 - (C) Before; in front of
 - (D) After; in front of
6. Solid medications are usually placed
 - (A) In pill organizers
 - (B) Directly on the resident's meal tray
 - (C) Directly on the resident's overbed table
 - (D) In a souffle cup
7. The MA should wear gloves
 - (A) When administering medications to the resident's eyes
 - (B) When transferring solid medications from a blister pack to a souffle cup
 - (C) When cleaning the medication station between uses
 - (D) When retrieving a medication that has fallen on the floor
8. What is the MA's responsibility when a resident refuses a medication?
 - (A) The MA should honor and document the refusal and report it to the nurse.
 - (B) The MA should skip the resident and try again later.
 - (C) The MA should ask another healthcare worker to help encourage the resident to take the medication.
 - (D) The MA should leave the medication with the resident in case he changes his mind.
9. Which of these examples is a way for an MA to practice person-centered care?
 - (A) Hiding medication in a resident's meal if he has difficulty swallowing medications alone
 - (B) Encouraging independence by allowing the resident to choose what time to take her medication
 - (C) Explaining the procedure to the resident every time it is performed
 - (D) Administering the medication as quickly as possible to avoid disrupting the resident's day
10. Which of the following can make swallowing oral medications easier for a resident?
 - (A) Swallowing all oral medications at once
 - (B) Sipping water first to moisten the mouth
 - (C) Allowing all oral medications to dissolve under the tongue
 - (D) Holding the medication between the front teeth to position it for easy swallowing
11. Which of the following statements is true regarding the MA's responsibilities during medication preparation?
 - (A) It is outside the scope of an MA's practice to prepare medications, but they are able to pass medications to the residents and document the procedures.
 - (B) The MA may prepare all nonprescription medications, but the nurse must prepare prescription medications.
 - (C) The MA must prepare and pass oral medications before beginning any other medication preparation.
 - (D) MAs must check the resident's MAR several times throughout the process of preparing and passing medications.

12. The MA should minimize _____ when preparing medications.
- (A) Distractions
 - (B) Safety checks
 - (C) The number of souffle cups used
 - (D) Side effects
13. What piece of information is always included on a prescription medication label?
- (A) The resident's weight
 - (B) The resident's allergies
 - (C) Medication strength by dose
 - (D) Medication strength by unit
14. The resident's _____ and the _____ on the medication label must match the resident's _____.
- (A) Room number; time; MAR
 - (B) First name; medication name; medication description
 - (C) Full name; medication name; MAR
 - (D) Known allergies; additional prescriptions; MAR
15. One souffle cup should contain
- (A) No more than one type of medication
 - (B) All solid oral medications to be administered to the same resident at one time
 - (C) All doses of one kind of medication the resident will need for 1 day
 - (D) Only solid medications that have been crushed or split
16. What is the difference between *medication strength* and a *medication dose*?
- (A) These two phrases have the same meaning.
 - (B) *Medication strength* refers to the amount of active ingredient per unit; *medication dose* refers to how much medication is to be administered at one time.
 - (C) *Medication strength* refers to the amount of time it takes for medication to become effective; *medication dose* refers to the amount of active ingredient per dose.
 - (D) *Medication strength* is documented in the MAR; *medication dose* is located only on the medication label.
17. What is the minimum number of safety checks an MA must perform while preparing medications?
- (A) Two
 - (B) Three
 - (C) Five
 - (D) Six
18. When transferring a medication from a blister pack to a souffle cup the MA should
- (A) Check for any damage to the pill by touching all pill surfaces
 - (B) Handle the pill by the edges
 - (C) Press the pill through the foil and into the souffle cup without touching it
 - (D) Use one souffle cup per pill
19. Which of the following might be used as an alternative to a medicine cup for measuring a dose of a liquid medication?
- (A) An oral syringe
 - (B) A souffle cup
 - (C) A spoon from the dining room
 - (D) A scale
20. The final check an MA should make of a resident's medication before passing it is when the medication is _____.
- (A) About to expire
 - (B) Replaced in storage
 - (C) Documented
 - (D) Placed in the dosage or souffle cup
21. Which of the following types of medications can be split (if indicated in the resident's MAR)?
- (A) A scored tablet
 - (B) An enteric-coated tablet
 - (C) A capsule
 - (D) A powdered medication
22. Which of the following is the first safety procedure an MA must perform prior to administering a medication of any type?
- (A) Donning gloves
 - (B) Documenting the procedure
 - (C) Identifying the resident
 - (D) Washing hands in front of the resident

23. Why should the MA ask the resident to state his name and date of birth rather than confirm them?
- (A) To check for cognitive decline
 - (B) To decrease the risk of an incorrect identification
 - (C) To learn the name of a new resident
 - (D) To make sure the resident's speech is not slurred
24. Residents should be in a(n) _____ position when taking oral medications.
- (A) Upright
 - (B) Supine
 - (C) Standing
 - (D) Dangling
25. If an MA arrives to administer a dose of medication requiring an empty stomach and learns the resident has just eaten she should
- (A) Leave the dose of medication with the resident and tell him what time is safe to take it
 - (B) Administer the dose as usual but document that the resident had a full stomach
 - (C) Not administer the drug, report it to the nurse, and document the reason for not administering the medication
 - (D) Ask the resident if he prefers to take the medication or wait
26. A buccal medication is placed _____ and a sublingual medication is placed _____.
- (A) Against the resident's cheek; beneath the resident's tongue
 - (B) Against the roof of the resident's mouth; beneath the resident's tongue
 - (C) Beneath the resident's tongue; far back on the resident's tongue
 - (D) Against the resident's cheek; far back on the resident's tongue
27. Which of the following is a medication type an MA would be allowed to administer?
- (A) Warm soak with additive
 - (B) Intramuscular antibiotic injection
 - (C) Intravenous medication with an established IV
 - (D) Vaccination
28. Which of the following types of medication may require a spacer?
- (A) Tincture
 - (B) Nebulizer
 - (C) Metered dose inhaler
 - (D) Nasal spray
29. What is the amount of time an MA should instruct the resident to hold his breath after using a metered dose inhaler?
- (A) 5 seconds
 - (B) 20 seconds
 - (C) 30 seconds
 - (D) As long as he can
30. Which of the following is the best example of an MA using proper communication skills while administering a medication?
- (A) The MA reminds a resident he does not like the taste of his liquid medication.
 - (B) The MA warns the resident that she will activate his inhaler on the count of 3.
 - (C) The MA jokes with the resident while preparing medications on her medication cart.
 - (D) The MA uses a friendly voice to insist that the resident take his medication.
31. Which of the following types of medications are easily administered with the resident lying down?
- (A) Eye drops and nasal spray
 - (B) Eye drops and nasal drops
 - (C) Eye drops and metered dose inhaler
 - (D) Nasal spray and nasal drops
32. When eye drops and ointment are both ordered, in which order should they be administered?
- (A) Instill eye drops and then immediately administer ointment.
 - (B) They should be alternated.
 - (C) Instill drops, wait 5 minutes, then administer ointment.
 - (D) Administer ointment, wait 5 minutes, then instill drops.
33. When administering medication by the ophthalmic route the MA must
- (A) Use alcohol-based hand rub
 - (B) Wash hands but not use gloves
 - (C) Wash hands and wear gloves
 - (D) Apply alcohol-based hand rub to gloved hands

34. How should the resident be positioned to receive medication administered by the otic route?
- (A) Lying back at a 45-degree angle
 - (B) In the upright position with the head tilted back
 - (C) With her head at the MA's eye level
 - (D) On her back with her head turned to the side
35. If an MA uses a tongue blade as an application device for administering a medication by the topical route he should
- (A) Sanitize it and store it with the resident's medication
 - (B) Discard it after use
 - (C) Wash it and replace it in the resident's bedside table
 - (D) Sanitize it to prepare it for use with another resident
36. What should an MA mark on a transdermal patch?
- (A) The MA's initials and the date and time the patch was applied
 - (B) The supervising nurse's initials and the date and time the patch was applied
 - (C) The MA's initials and the strength of the medication patch from the medication label
 - (D) The MA's initials and the date and time the patch should be removed
37. When is it necessary to handle a transdermal patch with gloves?
- (A) Gloves should be worn during the application, removal, and disposal processes.
 - (B) Gloves should be worn during the application process but are optional during removal.
 - (C) Gloves should be worn only if the resident has fragile or broken skin.
 - (D) A patch can be applied and removed with ungloved hands, but gloves should be donned for disposal.
38. If the resident _____ during administration of medication by the rectal route the MA should stop and report it to the nurse.
- (A) Becomes embarrassed
 - (B) Makes an insensitive joke
 - (C) Passes gas
 - (D) Reports pain
39. Administration of medication by which route requires the resident to be in the dorsal recumbent position?
- (A) Rectal
 - (B) Otic
 - (C) Oral
 - (D) Vaginal
40. How far inside the vagina of the resident should a medication applicator be inserted?
- (A) Most of the medication should be applied externally
 - (B) 1/2–1 inch
 - (C) 2–3 inches
 - (D) Until the applicator meets resistance
41. Which of the following statements is true of an MA's duties when assisting a resident with self-administered medications?
- (A) An MA may receive orders by telephone if the resident self-administers medication.
 - (B) An MA is not responsible for documenting medications that a resident self-administers.
 - (C) The MA may assist the resident by opening a medication container for the resident.
 - (D) The MA may convert medication dosages for the resident.
42. Which of the following could increase the chance of a medication error?
- (A) Documenting medication administration immediately after it is completed
 - (B) Documenting medication administration before it is completed
 - (C) Using an electronic system to document medication administration
 - (D) Checking a medication three times against the MAR before administering it

Chapter 9 Exam

1. Which of the following is an example of a medication error outside the MA's control?
 - (A) The MA gives the resident a drug with a name very close to the name of the ordered drug.
 - (B) The pharmacy begins to stock a different generic form of a commonly ordered medication.
 - (C) The pharmacist has incorrectly filled a prescription based on a medication order that was difficult to read.
 - (D) The MA administers an ordered medication she does not recognize the name of.
2. An MA can take an active role in helping to prevent medication errors by
 - (A) Memorizing which medications all of her assigned residents take
 - (B) Calling the pharmacist to confirm every medication order has been filled correctly
 - (C) Performing one medication check for each medication dosage
 - (D) Recognizing anything unusual about a medication order and asking for confirmation from the nurse
3. Which of the following would be an example of an error caused by an MA failing to follow instructions for use of a medication?
 - (A) The strength of a medication is not printed on the label.
 - (B) The MA did not correctly identify the resident.
 - (C) The MA looked up the wrong medication name in the medication reference book.
 - (D) The MA used a pill splitter on a large, enteric-coated tablet to make it easier for the resident to swallow.
4. _____ should be reported because even though an error was prevented, there may be a problem with a process that can be fixed to improve safety.
 - (A) Complaints
 - (B) Delays
 - (C) Near misses
 - (D) Refusals
5. Administering certain medications with a full glass of water, shaking certain liquid medications, and storing certain medications in the refrigerator are examples of
 - (A) Special instructions
 - (B) Common guidelines
 - (C) Facility policy
 - (D) Boxed warnings
6. Which of the following is an example of an error caused by an MA failing to follow facility policy?
 - (A) The MA measures a resident's blood pressure prior to administering a medication and reports the low blood pressure measurement to the nurse instead of administering the medication.
 - (B) The MA does not follow medication label instructions to shake a resident's inhaler and the resident does not receive the full benefit of a full dose of medication.
 - (C) The pharmacy replaces a resident's usual generic medication with the brand-name medication and the MA documents the new medication name.
 - (D) The MA leaves a medication cart unlocked and later discovers one blister pack of a resident's medication is missing.
7. Which of the following is the best way to document the quantity *fifteen milligrams* so as to reduce errors?
 - (A) 15.0 mg
 - (B) 15 mg
 - (C) 0.15 mg
 - (D) Fifteen mGs
8. How should an MA handle a question or concern about a medication from a resident's family member?
 - (A) The MA should suggest an online reference guide to the family member.
 - (B) The MA should assure the family member that the facility would not administer a medication that was not effective.
 - (C) The MA should answer any question the family member has as long as the resident has given permission.
 - (D) The MA should listen to the question or concern and report it to the supervisor.

9. Why is it important for an MA to report to the nurse if a resident does not feel a medication is working properly?
- (A) Because this will prevent an ineffective medication being administered to other residents
 - (B) Because residents who complain frequently should be identified
 - (C) Because reporting a resident's concerns may lead to a discovery of a medication error such as an improperly filled prescription or dosage mistake
 - (D) Because reporting a resident's concerns shows that the MA has a proper understanding of facility policy
10. Which of the following is true about medication errors involving PRN medications?
- (A) Because PRN medications are often over-the-counter drugs, there is no need to report medication errors involving them.
 - (B) Errors with all medications, including PRN medications, should be documented and reported immediately.
 - (C) Errors with PRN medications do not need to be reported unless they cause a severe reaction.
 - (D) Errors with PRN medications only need to be reported if the error causes an overdose.
11. How are an MA's ethics related to resident safety?
- (A) An ethical MA does not make errors with a resident's medication.
 - (B) An ethical MA values being on a team and can be counted on to cover for care team members' mistakes.
 - (C) An ethical MA is more concerned with resident safety than possible disciplinary action for making an error.
 - (D) An ethical MA is not afraid to discipline coworkers for endangering a resident by making a medication error.
12. One role of the FDA is to _____ new medications.
- (A) Develop
 - (B) Test
 - (C) Approve
 - (D) Fund
13. How is the CMS (Centers for Medicare & Medicaid Services) involved in ensuring medication safety in long-term care?
- (A) The Joint Commission hired CMS to enforce all safety measures and control facility budgets.
 - (B) The CMS makes surveys of the types and severity of medication errors in facilities receiving CMS funding.
 - (C) The CMS makes surveys of the types and severity of medication errors in private facilities.
 - (D) The CMS is not involved in medication safety in any way.

Chapter 10 Exam

1. Which of the following is true of skin?
 - (A) Skin is a sense organ.
 - (B) Skin becomes thicker with age.
 - (C) Sweat glands in the skin help heat the body.
 - (D) Subcutaneous tissue makes up the middle layer of skin.
2. Which of the following medication classes includes drugs commonly used to treat cellulitis?
 - (A) Antifungals
 - (B) Antibiotics
 - (C) Antivirals
 - (D) Antipruritics
3. A scabicide is a medication that kills
 - (A) Bacteria
 - (B) Mites
 - (C) Fungus
 - (D) Scabs
4. Which of the following is true about medications used to treat conditions of the integumentary system?
 - (A) Integumentary system conditions may be treated with a combination of topical and oral medications.
 - (B) Oral medications are ineffective against conditions of the integumentary system.
 - (C) Medications used to treat conditions of the integumentary system rarely require a prescription.
 - (D) Open wounds require topical medications while closed wounds require oral medications.
5. Which of the following should be immediately reported to the nurse if a resident is taking antifungal medication?
 - (A) Body odor
 - (B) Photosensitivity
 - (C) Yellowish eyes
 - (D) Mild itching
6. Prescription topical corticosteroids work by _____ and may have systemic side effects such as _____.
 - (A) Reducing circulation; easy bruising
 - (B) Reducing inflammation; lowered infection resistance
 - (C) Reducing inflammation; skin redness
 - (D) Reducing inflammation; nausea and vomiting
7. One cause of cellulitis may be
 - (A) Aging
 - (B) MRSA
 - (C) Fungus
 - (D) Mites
8. Fingernail discoloration may be a sign of
 - (A) Ringworm
 - (B) Cellulitis
 - (C) Psoriasis
 - (D) Fungal infection
9. Silver sulfadiazine may be used to reduce the risk of infection in residents with
 - (A) Eczema
 - (B) Burns
 - (C) MRSA
 - (D) Pediculosis
10. Psoriasis is caused by a(n) _____ and may be treated with _____.
 - (A) Fungal infection; miconazole
 - (B) Sunburn; aloe gel
 - (C) Sunburn; vitamin D analog
 - (D) Autoimmune condition; immunosuppressant drug
11. Which of the following conditions may be treated with medication in powder form?
 - (A) Fungal infection
 - (B) Cellulitis
 - (C) Psoriasis
 - (D) Atopic dermatitis

Chapter 11 Exam

1. Which of the following disorders of the musculoskeletal system is caused by aging?
(A) Gout
(B) Closed fractures
(C) Osteomyelitis
(D) Osteoporosis
2. There are _____ bones in the human skeleton.
(A) 166
(B) 206
(C) 236
(D) 246
3. Which of the following is true about rheumatoid arthritis and osteoarthritis?
(A) Rheumatoid arthritis is caused by injury and osteoarthritis is caused by aging.
(B) Rheumatoid arthritis causes stiffness and osteoarthritis causes bones to become porous.
(C) Rheumatoid arthritis is the reversible form of arthritis; osteoarthritis is the permanent form.
(D) Rheumatoid arthritis is caused by an autoimmune illness while osteoarthritis is caused by aging and/or injury.
4. Analgesic medications reduce _____.
(A) Clotting
(B) Pain
(C) Bone loss
(D) Uric acid
5. Which of the following sets is an example of one common OTC NSAID and one common prescription NSAID?
(A) Advil and Celebrex
(B) Aleve and morphine
(C) Zylprim and Advil
(D) Tylenol and Aleve
6. Oral NSAIDs may be contraindicated for residents with _____ over concerns about possible _____.
(A) Degenerative joint disease; loss of bone density
(B) Sprains; swelling
(C) Heart failure; myocardial infarction
(D) Liver impairment; stomach ulcers
7. What is true of topical NSAIDs?
(A) Topical NSAIDs are more effective for relieving gout pain than osteoarthritis pain.
(B) Topical NSAIDs are effective for some conditions and have fewer side effects than oral NSAIDs.
(C) Topical NSAIDs are effective for all conditions and have more side effects than oral NSAIDs.
(D) There is no such thing as a topical NSAID.
8. The gout drug colchicine has a(n) _____ therapeutic index which means the MA should be especially aware of signs of _____.
(A) Narrow; toxicity
(B) Narrow; ineffectiveness
(C) Accelerated; withdrawal
(D) Strong; toxicity
9. What do bisphosphonates do?
(A) Reduce uric acid
(B) Prevent bone resorption
(C) Decrease muscle tone
(D) Relieve pain
10. What is a possible cause of osteomyelitis?
(A) Bone loss
(B) Gout
(C) Pressure injuries
(D) Autoimmune illness
11. Which of the following musculoskeletal conditions is commonly treated with skeletal muscle relaxants?
(A) Fractures
(B) Back pain
(C) Osteoporosis
(D) Strains

Chapter 12 Exam

1. How is a transient ischemic attack (TIA) related to a stroke?
 - (A) *Transient ischemic attack* is the medical term for a stroke.
 - (B) A stroke occurs in the brain and TIA can occur elsewhere in the body such as a major vessel.
 - (C) TIA is a temporary loss of blood supply to part of the brain while a stroke is a longer-term blockage or loss of blood supply to part of the brain.
 - (D) TIA is the name for a stroke that has no symptoms after the first 24 hours.
2. The B.E.F.A.S.T. acronym can be used to help remember the signs of
 - (A) A stroke
 - (B) An overdose
 - (C) A seizure
 - (D) Schizophrenia
3. Which of the following is true regarding epilepsy?
 - (A) Epilepsy is primarily controlled using anti-platelet agents.
 - (B) Epilepsy may have known or unknown causes.
 - (C) Epilepsy is caused by TIA.
 - (D) Epilepsy usually exists along with myasthenia gravis.
4. Which of the following is an antiepileptic drug from the benzodiazepine drug class?
 - (A) Phenobarbital
 - (B) Lamotrigine
 - (C) Diazepam
 - (D) Aspirin
5. Multiple sclerosis has no cure, but treatments include _____, which reduce nerve inflammation, and _____, which regulate the immune system.
 - (A) Corticosteroids; antispasmodics
 - (B) NSAIDs; immunosuppressants
 - (C) Corticosteroids; immunomodulators
 - (D) NSAIDs; antispasmodics
6. For which of the following nervous system conditions might a resident be prescribed a cholinesterase inhibitor?
 - (A) Orthostatic hypotension
 - (B) Neuropathic pain
 - (C) Epilepsy
 - (D) Myasthenia gravis
7. This class of drugs can help ease the tremors and involuntary muscle movements related to Parkinson disease.
 - (A) Anticholinergic agents
 - (B) ACE inhibitors
 - (C) Anxiolytic agents
 - (D) Opioid analgesics
8. Which of the following statements is true regarding medications and Alzheimer disease?
 - (A) There are currently no medications that effectively treat the symptoms of Alzheimer disease.
 - (B) It is unethical to administer medications to residents with Alzheimer disease as they may not be able to understand their right to refuse.
 - (C) NMDA receptor agents block an over-produced brain chemical and may lessen symptoms in residents with Alzheimer disease.
 - (D) Immunosuppressants can slow the progression of Alzheimer disease and are often used in the first stage of the disease.
9. How is neuropathic pain different from musculoskeletal pain?
 - (A) Neuropathic pain is constant and musculoskeletal pain is transient.
 - (B) Neuropathic pain is treated systemically and musculoskeletal pain is treated locally.
 - (C) Neuropathic pain is a type of phantom pain and musculoskeletal pain is a type of physical pain.
 - (D) Musculoskeletal pain is caused by injury to muscles, bones, or joints while neuropathic pain is caused by a disease or damage to the nervous system.

10. Opioids are usually the _____ for treating neuropathic pain.
- (A) First resort
 - (B) Last resort
 - (C) Safest
 - (D) Least effective
11. Which of the following medications is often used to treat anxiety?
- (A) A mood stabilizer such as lithium
 - (B) An antidepressant such as trazodone
 - (C) An antipsychotic such as haloperidol
 - (D) An SNRI such as fluoxetine
12. In a resident taking lithium for bipolar disorder, the resident's _____ might need to be monitored.
- (A) Salt intake
 - (B) Fluid intake
 - (C) Fluid output
 - (D) Blood sugar
13. Which of the following statements is true regarding medications for psychotic disorders?
- (A) Medications are rarely effective in the treatment of psychotic disorders.
 - (B) Anxiolytic agents are the most common treatment for psychotic disorders but they have a narrow therapeutic index.
 - (C) EPS is a group of symptoms often caused by second-generation antipsychotics.
 - (D) Both first- and second-generation antipsychotic medications have a boxed warning about increased death risk to elderly patients with dementia.
14. Which of the following statements is true regarding the treatment of glaucoma?
- (A) Glaucoma is caused by dry eyes and is treated with preservative-containing artificial tears.
 - (B) Glaucoma is a bacterial condition that is treated with ophthalmic antibiotics in drop or ointment form.
 - (C) NSAIDs can relieve the discomfort of glaucoma until it can be surgically corrected.
 - (D) Alpha-2 adrenergic agonists, which reduce the production of eye fluid, are an effective treatment for glaucoma.
15. Which of the following statements is true regarding the treatment of cataracts?
- (A) Use of NSAIDs can delay the need for surgery in residents with cataracts.
 - (B) NSAIDs may be ordered after the removal of a cataract to reduce post-surgery inflammation.
 - (C) Cataracts develop because of the overuse of the ophthalmic antibiotic ciprofloxacin in the treatment of chronic conjunctivitis.
 - (D) NSAIDs are regularly ordered for residents with cataracts to keep the condition from progressing to macular degeneration.
16. What is the purpose of a cerumenolytic?
- (A) To cure infection in the middle ear
 - (B) To dry liquid trapped in the inner ear
 - (C) To soften earwax for removal
 - (D) To reduce swelling in the ear

Chapter 13 Exam

1. How does a diuretic work in the body to lower blood pressure?
(A) By relaxing the heart muscle
(B) By increasing the amount of fluid in the body
(C) By lowering the amount of fluid in the body
(D) By removing potassium from the body
2. How does a calcium channel blocker work in the body to lower blood pressure?
(A) By relaxing the heart and blood vessels
(B) By reducing a person's stress and anxiety
(C) By lowering the amount of calcium in the bones
(D) By lowering the amount of fluid in the body
3. Which of the following blood pressure medication classes can cause a life-threatening adverse reaction called angioedema?
(A) Calcium channel blockers
(B) Beta-blockers
(C) ACE inhibitors
(D) Diuretics
4. Which of the following pairs the drug class with the correct suffix used in generic medication names?
(A) ACE inhibitors; *-pine*
(B) ARBs; *-pril*
(C) Beta-blockers; *-olol*
(D) Calcium channel blockers; *-or*
5. A vasodilator works by
(A) Widening blood vessels
(B) Narrowing blood vessels
(C) Controlling the chemicals that increase blood pressure
(D) Controlling inflammation
6. The suffix *-sartan* applies to generic medications belonging to which of the following classes?
(A) Calcium channel blockers
(B) Beta-blockers
(C) ACE inhibitors
(D) ARBs
7. Which of the following can cause a buildup inside the arteries causing atherosclerosis?
(A) LDL cholesterol
(B) Potassium
(C) HDL cholesterol
(D) Insulin
8. Diuretics are typically administered _____ because _____.
(A) in the morning; frequent urination can disrupt sleep
(B) in the evening; they work overnight to relieve symptoms
(C) after each meal; they work to prevent salts from being absorbed
(D) with food; they can irritate the stomach
9. What is the purpose of a statin?
(A) It relaxes the heart muscle.
(B) It disrupts fat absorption.
(C) It reduces cholesterol levels.
(D) It thins the blood.
10. Which of the following interferes with several drugs used to treat cardiovascular conditions?
(A) Dairy products
(B) Foods high in potassium
(C) Leafy greens
(D) Grapefruit and grapefruit juice
11. Which of the following is a vitamin used to decrease triglycerides?
(A) Potassium
(B) Vitamin D
(C) Niacin
(D) Vitamin C
12. Which of the following statements about coronary artery disease (CAD) is true?
(A) Aging is the cause of CAD.
(B) CAD is often treated using the same medications as other cardiac conditions.
(C) CAD is a general term used to describe all conditions related to the heart.
(D) Antiplatelet agents are not effective in the treatment of CAD.

13. Which of the following OTC medications is used as an antiplatelet agent to treat CAD?
 - (A) Advil
 - (B) Aspirin
 - (C) Nitroglycerin
 - (D) Niacin
14. Nitroglycerin is commonly used in which of the following forms?
 - (A) Oral tablet, sublingual capsule, and nasal spray
 - (B) Ointment, sublingual tablet, and transdermal patch
 - (C) Sublingual tablet, syrup, and transdermal patch
 - (D) Transdermal patch, sublingual spray, sublingual tablet
15. Which of the following is a sign of a myocardial infarction that should be reported immediately?
 - (A) Sudden weight gain
 - (B) Severe chest pain
 - (C) Elevated blood pressure
 - (D) Sudden muscle spasms in the left arm
16. Heart failure (HF) refers to which of the following?
 - (A) Any viral, bacterial, or structural damage to the heart
 - (B) The age-related steady deterioration of the heart muscle over time
 - (C) A genetic disorder that causes the heart to degenerate
 - (D) The failure of one or both sides of the heart to pump blood efficiently
17. Which of the following is a medication commonly ordered to help regulate heart rhythm?
 - (A) Carvedilol, a beta-blocker
 - (B) Clopidogrel, an antiplatelet agent
 - (C) BiDil, an arterial dilator/nitrate medication
 - (D) Gemfibrozil, a fibrate
18. Residents taking oral iron should avoid taking it with which of the following?
 - (A) Thickened liquids
 - (B) High-fiber foods
 - (C) Vitamin C
 - (D) Any food or beverage other than water
19. Deep vein thrombosis is treated primarily with
 - (A) Increased exercise and compression bandages or stockings
 - (B) Antibiotic medication
 - (C) Anticoagulant medication
 - (D) A combination of iron and vitamin C
20. In residents taking anticoagulant medications, _____ may be particularly dangerous because of increased _____ risks.
 - (A) Iron; constipation
 - (B) Strokes; seizure
 - (C) Mobility; fall
 - (D) Accidents; bleeding

Chapter 14 Exam

1. Which of the following conditions may be treated with the same corticosteroid, administered by different routes?
(A) Rhinitis and pneumonia
(B) Rhinitis and asthma
(C) Sinusitis and pleurisy
(D) Asthma and influenza
2. What causes coughing and wheezing in residents with asthma?
(A) Reaction to inhaled medicines
(B) Bacteria in the respiratory tract
(C) Inability to get oxygen into the lungs
(D) Air trapped in the lungs
3. Which of the following is true about oral medications and asthma?
(A) There are no oral medications that are effective treatments for asthma.
(B) Oral corticosteroids may be used short-term to treat asthma flare-ups.
(C) Oral medications for asthma should be used in a nebulizer.
(D) Oral medications for asthma are all fast-acting rescue medications.
4. A resident receiving a dose of nasal spray should be instructed to
(A) Rinse his mouth after use
(B) Lie flat for administration
(C) Blow his nose first
(D) Pinch the bridge of his nose for 5 minutes
5. Which of the following is a short-acting bronchodilator used for an asthma attack?
(A) Montelukast
(B) Fluticasone
(C) Albuterol
(D) Dexamethasone
6. Which of the following statements is true regarding medications used to treat influenza?
(A) Influenza can be prevented with a flu vaccine and symptoms can be eased with medications, but there are no effective treatments to shorten the duration or severity of influenza.
(B) Antitussive agents such as codeine can shorten the duration and severity of influenza.
(C) Antiviral agents such as oseltamivir can shorten the duration and severity of influenza.
(D) Anti-inflammatory agents such as roflumilast can stop influenza from developing into pneumonia.
7. Why might it be important for an MA to know whether a resident contracted pneumonia at a hospital?
(A) Hospital-acquired pneumonia must be tracked separately for reporting purposes per the CMS.
(B) Hospital-acquired pneumonia is treated with specific antibiotics.
(C) It is outside the scope of an MA's practice to administer medications to residents with non-hospital-acquired pneumonia.
(D) Residents with hospital-acquired pneumonia may not receive medications containing acetaminophen.
8. Which of these statements about medications for emphysema is true?
(A) Short-term anti-inflammatory agents such as prednisone are commonly ordered to cure emphysema.
(B) Antibiotic agents such as doxycycline are commonly ordered to cure emphysema.
(C) Mucolytics such as inhaled mannitol are commonly ordered to cure emphysema.
(D) Inhaled long-acting bronchodilators such as glycopyrrolate can treat but not cure emphysema.
9. Pleurisy can be caused by
(A) Pneumonia or influenza
(B) Overuse of antibiotics
(C) Overuse of NSAIDs
(D) Anticoagulants used to treat pulmonary embolism

10. Which part of a resident's treatment for a pulmonary embolism is most likely to be delegated to an MA?
 - (A) Diagnosis of a blood clot
 - (B) Administration of an injected anticoagulant medication
 - (C) Assisting with the resident's active assisted range of motion exercises
 - (D) Administration of oral anticoagulant medication
11. Which of the following is used in both the active phase and the continuation phase of treatment for tuberculosis?
 - (A) Rifampin
 - (B) Pyrazinamide
 - (C) Indacaterol
 - (D) Erythromycin
12. What must MAs assist residents to do after administering inhaled corticosteroid medications?
 - (A) Hold breath for 15 seconds
 - (B) Use an incentive spirometer
 - (C) Rinse the mouth
 - (D) Perform deep breathing exercises
13. Antiviral medication for COVID-19 can
 - (A) Prevent the spread of the virus at a facility
 - (B) Reduce the risk of severe illness
 - (C) Prevent a resident from getting COVID-19 again in the next several months
 - (D) Stop the virus from having any effect

Chapter 15 Exam

1. Which of the following statements regarding urinary incontinence is true?
 - (A) Incontinence is not a normal part of aging.
 - (B) Incontinence is a normal part of aging.
 - (C) Incontinence can only be treated surgically.
 - (D) Incontinence must be treated with diet and exercise.
2. Urinary incontinence can sometimes be successfully treated by the _____ medication duloxetine.
 - (A) Beta-blocker
 - (B) Antidepressant
 - (C) Antibiotic
 - (D) Analgesic
3. Which of the following hormones in tablet form may be inserted vaginally to treat incontinence?
 - (A) Prolactin
 - (B) Testosterone
 - (C) Estrogen
 - (D) Cortisol
4. Which of the following drugs can be used to treat both males and females with urinary retention?
 - (A) Alfuzosin
 - (B) Tamsulosin
 - (C) Doxazosin
 - (D) Estrogen
5. Which of the following are common side effects of cholinergic agents used to treat urinary retention?
 - (A) Insomnia and dizziness
 - (B) Orthostatic hypertension
 - (C) Constipation and dry mouth
 - (D) Diarrhea and abdominal cramps
6. Which of the following may be administered to allow the body to pass small kidney stones?
 - (A) Estrogen by the vaginal route
 - (B) Alpha-blockers by the oral route
 - (C) Analgesics by the transdermal route
 - (D) Diuretics by the oral route
7. Uncomplicated UTIs may be infections of the _____ and/or _____ and are treated with oral antibiotics. Complicated UTIs require _____ antibiotics.
 - (A) Urethra; kidneys; stronger
 - (B) Kidneys; bladder; systemic
 - (C) Kidneys; bladder; intravenous
 - (D) Urethra; bladder; multiple
8. Which of the following may be ordered for an uncomplicated UTI in a resident with pyelonephritis?
 - (A) Levofloxacin
 - (B) Nitrofurantoin monohydrate
 - (C) Vancomycin
 - (D) Terazosin
9. A resident with chronic kidney disease may take which types of medications usually associated with another body system?
 - (A) Analgesics and anti-gout medications
 - (B) Antidepressants and anxiolytics
 - (C) ACE inhibitors and statins
 - (D) Inhaled corticosteroids and bronchodilators
10. Chronic kidney disease is often associated with which of the following conditions?
 - (A) Pulmonary embolism
 - (B) Hypotension
 - (C) Myocardial infarction
 - (D) Hypertension
11. Which of the following is true of kidney disease and MA responsibilities?
 - (A) MAs will supervise dialysis for residents with CKD.
 - (B) Because performing dialysis is beyond an MA's scope of practice, MAs will have no responsibilities for the care of residents with CKD.
 - (C) MAs will be responsible for ensuring that residents with CKD drink extra water during every medication pass.
 - (D) MAs will likely administer medications for conditions of other body systems affected by kidney failure.

12. Which of the following is a common symptom of UTI in elderly people?
- (A) Headache
 - (B) Confusion
 - (C) High blood pressure
 - (D) Chest pains

Chapter 16 Exam

1. A resident experiencing discomfort due to stomatitis may be given medication in the form of a(n) _____.
(A) Suppository
(B) Inhaler
(C) Lozenge
(D) Oral capsule
2. An important fact for an MA to know about hydrogen peroxide is
(A) It must be administered prior to brushing teeth.
(B) It must be diluted with water.
(C) It must be sprayed directly onto the affected areas of the mouth.
(D) It works by numbing the mouth.
3. An MA is preparing medications for two residents with stomatitis. A topical corticosteroid applied with a swab is ordered for one. A corticosteroid in oral rinse form is ordered for the other. Which of the following is likely true?
(A) The resident with the medication order for the swabbed form may have a cognitive impairment.
(B) The resident with the medication order for the rinse form wears dentures.
(C) The resident receiving the medication order for the oral rinse is likely to be unconscious.
(D) There is an error on one resident's MAR as corticosteroids to treat stomatitis only come in rinse form.
4. Which of the following may be ordered to treat a fungal infection in the mouth?
(A) Listerine
(B) Oral penicillin
(C) Nystatin rinse
(D) Tetracycline rinse
5. Which of the following is an H2 blocker?
(A) Famotidine
(B) Lansoprazole
(C) Misoprostol
(D) Chlorhexidine
6. If an MA sees an order for a proton pump inhibitor such as esomeprazole or lansoprazole, he should make sure to administer the medication _____.
(A) Only when the resident requests it
(B) 30–60 minutes before a meal
(C) Before bed
(D) With a stool softener
7. Which three conditions of the gastrointestinal system may sometimes be treated with the same medications?
(A) Peptic ulcers, GERD, and gastritis
(B) GERD, gastritis, and ulcerative gingivitis
(C) Gastritis, cirrhosis, and GERD
(D) Gastritis, pancreatitis, and peptic ulcers
8. What is the purpose of an antiemetic medication?
(A) Antiemetics are for rehydration.
(B) Antiemetics slow the movement of the gastrointestinal tract.
(C) Antiemetics slow or prevent vomiting.
(D) Antiemetics provide beneficial bacteria to the gastrointestinal tract.
9. Which of the following are examples of antimotility agents used to treat diarrhea?
(A) Bismuth subsalicylate and loperamide
(B) Diphenhydramine and dimenhydrinate
(C) Bismuth subsalicylate and diphenhydramine
(D) Loperamide and scopolamine
10. How can treatment for constipation also work to ease hemorrhoid discomfort?
(A) Laxatives contain agents that numb the rectal area.
(B) Treating constipation can reduce straining during bowel movements.
(C) The ingredient in powdered fiber supplements (psyllium) reduces rectal itching.
(D) Laxative suppositories decrease the size of hemorrhoids.
11. Which of the following is a possible side effect of laxatives used to treat constipation?
(A) Drowsiness
(B) Dizziness
(C) Delirium
(D) Diarrhea

12. IBS with diarrhea and/or constipation is caused by
- (A) Miscommunication between the lymphatic system and the gastrointestinal tract
 - (B) The spore-forming bacterium *C. difficile*
 - (C) Miscommunication between the nervous system and the gastrointestinal tract
 - (D) Surgery or trauma to the lower gastrointestinal tract
13. IBS may be treated with the _____ agent dicyclomine while IBD may be treated with the immunosuppressant medication _____.
- (A) Antimotility; mesalamine
 - (B) Antibiotic; rifaximin
 - (C) Probiotic; methotrexate
 - (D) Antispasmodic; tofacitinib
14. An oral antibiotic such as _____ may be ordered to treat mild cases of _____, which is a type of inflammation in the colon.
- (A) Levofloxacin; diverticulitis
 - (B) Levofloxacin; Crohn disease
 - (C) Amoxicillin/clavulanate; ruptured diverticula
 - (D) Ciprofloxacin; cirrhosis
15. The antiviral agents lamivudine, telbivudine, and tenofovir are all medications that treat hepatitis and are administered by the _____ route.
- (A) Subcutaneous
 - (B) IV
 - (C) Oral
 - (D) Rectal
16. Vaccinations for which of the following diseases can help protect the liver from further damage in a resident with cirrhosis?
- (A) Hepatitis A and C; tetanus
 - (B) Hepatitis B and C; influenza
 - (C) Hepatitis A, influenza, and pneumococcal pneumonia
 - (D) Hepatitis A and B; pneumococcal pneumonia
17. Treatment for gallstones in an older person who is not well enough for surgery may include using _____ to dissolve the gallstones over a _____ period of time.
- (A) Ibuprofen; long
 - (B) Ursodiol; long
 - (C) Morphine; short
 - (D) Ursodiol; short
18. Although pancreatitis is usually treated in a hospital, medication aides should know that certain _____ such as _____ should not be used to treat the pain associated with pancreatitis due to increased bleeding risks.
- (A) anticoagulants; warfarin
 - (B) analgesics; acetaminophen
 - (C) analgesics; morphine
 - (D) analgesics; NSAIDs
19. Which of the following is a side effect that may be observed in a resident taking pancreatic enzymes?
- (A) Increased blood pressure
 - (B) Gastrointestinal discomfort
 - (C) Ulcers
 - (D) Anaphylaxis

Chapter 17 Exam

1. Which of the following best explains how insulin works in the body?
 - (A) Insulin breaks down glucose in the bloodstream to create body fat.
 - (B) Insulin moves glucose from the cells into the bloodstream to provide energy for the body.
 - (C) Insulin regulates the speed of cell turnover in the body.
 - (D) Insulin moves glucose from the bloodstream into the cells to provide energy for the body.
2. The inhaled form of insulin has a boxed warning for patients who have which of the following conditions?
 - (A) Chronic lung disease
 - (B) Hypothyroidism
 - (C) Type 2 diabetes
 - (D) Chronic renal failure
3. Which of the following disorders of the endocrine system may be treated with use of an insulin pump?
 - (A) Hypothyroidism
 - (B) Type 1 diabetes
 - (C) Prediabetes
 - (D) Well-controlled type 2 diabetes
4. What type of regular testing must residents with diabetes have performed?
 - (A) Liver enzyme tests
 - (B) Blood coagulation test
 - (C) Blood glucose test
 - (D) Blood lipid profile
5. The antidiabetic agents repaglinide and glimepiride work by
 - (A) Increasing sensitivity to insulin
 - (B) Stimulating the release of insulin
 - (C) Decreasing production and absorption of glucose
 - (D) Releasing a steady amount of insulin into the bloodstream
6. Which of the following is an antidiabetic agent that may cause acute pancreatitis?
 - (A) Sitagliptin
 - (B) Levothyroxine
 - (C) Metformin
 - (D) Potassium iodide
7. Which of the following is a sign of liver damage that should be immediately reported in residents taking pioglitazone or rosiglitazone?
 - (A) Increased urination
 - (B) Dark urine
 - (C) Body aches
 - (D) Weight gain
8. Which of the following guidelines is important to follow when administering oral antidiabetic medications?
 - (A) Do not provide any food or beverages not listed in the MAR.
 - (B) Check the resident's blood sugar before administering the medications.
 - (C) Hold antidiabetic medications if the resident is not showing signs and symptoms of diabetic ketoacidosis.
 - (D) Only administer these medications if the nurse has assessed the resident and approved it.
9. What is the MA's role when a resident is experiencing insulin reaction?
 - (A) The MA should immediately provide something sugary such as candy or juice.
 - (B) The MA should request that the nurse give a stat order for glucose tablets.
 - (C) The MA should report observations to the nurse immediately.
 - (D) The MA should encourage the resident to vomit.
10. Which of the following is a common treatment for hyperthyroidism?
 - (A) Insulin
 - (B) Niacin
 - (C) Estrogen
 - (D) Potassium iodide
11. What is important for an MA to know about thyroid hormone replacement medications?
 - (A) They must be administered on an empty stomach.
 - (B) They must be administered 30 minutes after a meal.
 - (C) They may cause acute pancreatitis.
 - (D) They are an effective treatment for hyperthyroidism.

Chapter 18 Exam

1. Difficulty starting the flow of urine may be a sign of
 - (A) Erectile dysfunction
 - (B) Vaginitis
 - (C) Benign prostatic hyperplasia
 - (D) Yeast infection
2. The medication _____ can be used to treat BPH or erectile dysfunction.
 - (A) Tadalafil
 - (B) Terazosin
 - (C) Finasteride
 - (D) Imiquimod
3. Which of the following medications should not be handled by MAs who are pregnant or may become pregnant?
 - (A) Clindamycin
 - (B) Tadalafil
 - (C) Doxazosin
 - (D) Finasteride
4. The alpha-blocker _____ may be used in the treatment of BPH and should be administered 30 minutes after _____ every day.
 - (A) Tamsulosin; each meal
 - (B) Tamsulosin; the same meal
 - (C) Vardenafil; each meal
 - (D) Vardenafil; the first meal
5. The medication ordered for _____ will depend on the _____ of the infection.
 - (A) Genital herpes; cause
 - (B) Genital HPV infection; duration
 - (C) Vaginitis; cause
 - (D) Vaginitis; duration
6. Medications for which condition cannot be administered with nitrates?
 - (A) Vaginitis
 - (B) Genital herpes
 - (C) Erectile dysfunction
 - (D) Candidiasis
7. For which of the following conditions might a female resident be prescribed hormone replacements by the vaginal route?
 - (A) Genital HPV infection
 - (B) Bacterial vaginitis
 - (C) Fungal vaginitis
 - (D) Vaginal atrophy
8. Which of the following sexually transmitted infections are bacterial and treated with antibiotics?
 - (A) Chlamydia and gonorrhea
 - (B) HIV and genital HPV
 - (C) Genital herpes and genital HPV infection
 - (D) Gonorrhea and erectile dysfunction
9. Valacyclovir may be ordered to treat which of the following?
 - (A) Genital HPV infection
 - (B) Syphilis
 - (C) Genital herpes
 - (D) HIV
10. _____, a condition of the musculoskeletal system, may be treated with _____.
 - (A) Rheumatoid arthritis; estrogen replacement therapy
 - (B) Osteomyelitis; immunosuppressant
 - (C) Osteoporosis; immunosuppressant
 - (D) Osteoporosis; estrogen replacement therapy
11. The same alpha-blockers used to treat _____ are also commonly used to treat _____.
 - (A) Erectile dysfunction; BPH
 - (B) Urinary retention; BPH
 - (C) BPH; cystitis
 - (D) BPH; CRF

Chapter 19 Exam

1. To which other body systems is the lymphatic system most closely related?
 - (A) The immune and circulatory systems
 - (B) The immune and respiratory systems
 - (C) The circulatory and respiratory systems
 - (D) The circulatory and integumentary systems
2. Why does it become increasingly difficult to treat AIDS patients for infections such as pneumonia and TB?
 - (A) Because AIDS causes an allergic response to antibiotics
 - (B) Because AIDS patients get more infections, they can become resistant to many antibiotics.
 - (C) Because antiretroviral therapies to treat AIDS decrease the effectiveness of certain antibiotics
 - (D) Because AIDS causes the breakdown of lymph vessels, preventing them from distributing antibiotics throughout the body
3. Opportunistic infections occur
 - (A) As soon as a person is exposed to HIV
 - (B) During the first (acute) phase of HIV infection
 - (C) During the second (chronic) phase of HIV infection
 - (D) During the third stage of infection (AIDS)
4. Certain drugs taken as part of antiretroviral therapy interact with which medication?
 - (A) The beta-blocker metoprolol
 - (B) The statin simvastatin
 - (C) The bronchodilator salmeterol
 - (D) The anticoagulant warfarin
5. Observing and reporting changes to the emotional state of a resident taking the medication efavirenz may lead to the prevention of _____.
 - (A) Suicide
 - (B) The progression of HIV to AIDS
 - (C) Liver damage
 - (D) Insomnia
6. In residents with cancer, MAs will be responsible for administering medications such as _____ to help relieve the _____ of cancer treatments.
 - (A) Analgesics; fatigue
 - (B) Antiemetics; pain
 - (C) Antiemetics; side effects
 - (D) Antiemetics; hair loss
7. Why are osteoarthritis and rheumatoid arthritis treated with different types of medications?
 - (A) *Osteoarthritis* is what the early stage of the disease is called before it progresses to rheumatoid arthritis, which requires stronger medications.
 - (B) Although arthritis starts in the musculoskeletal system, when it spreads to the immune system it becomes rheumatoid arthritis, requiring different classes of analgesics.
 - (C) Osteoarthritis cannot be treated effectively with disease-modifying antirheumatic drugs (DMARDs) because it does not involve the immune system, while rheumatoid arthritis is an autoimmune disorder that is slowed by the use of DMARDs.
 - (D) Corticosteroids can effectively ease the symptoms associated with osteoarthritis because it is caused by inflammation in the joints, but because rheumatoid arthritis does not cause inflammation, corticosteroids are an ineffective treatment for that condition.
8. Drugs in this medication class used to treat rheumatoid arthritis may be tapered down rather than stopped immediately:
 - (A) DMARDs
 - (B) Corticosteroids
 - (C) NSAIDs
 - (D) bDMARDs
9. For which condition might a resident be prescribed the antimalarial agent hydroxychloroquine?
 - (A) Lupus
 - (B) AIDS
 - (C) Influenza
 - (D) Rheumatoid arthritis

10. It is common for residents taking immunosuppressant medications to experience an increase in _____.
 - (A) Appetite
 - (B) Infections
 - (C) Insomnia
 - (D) Pain
11. Which of the following may be used to shorten the duration of a shingles outbreak?
 - (A) Amitriptyline
 - (B) Pregabalin
 - (C) Morphine
 - (D) Acyclovir
12. Which of the following medication classes may be effective in treating symptoms of postherpetic neuralgia?
 - (A) Dopamine replacement agents and antibiotics
 - (B) Immunosuppressants and immunomodulatory agents
 - (C) SNRIs and tricyclic antidepressants
 - (D) Tricyclic antidepressants and anticonvulsants
13. Residents over the age of 50 may be vaccinated for which of the following immune system conditions?
 - (A) Rheumatoid arthritis
 - (B) Lupus
 - (C) Herpes zoster
 - (D) Genital herpes